Impostor hunt class handout

Name: _____

Read the question and both answers. Choose exactly one answer, A or B, as the AI impostor. Do not vote on your own submission. Record your confidence and reason before the reveal. Please do not follow source links until voting is complete.

Round 1 - College and personal growth

Question: What does “college” mean to you at this point in your life? What do you hope college will help you discover or become?

Who or what has influenced your definition of college up to this point? What assumptions or expectations are you bringing into college? Which do you feel confident about, and which feels uncertain or unclear?

Do you have any expectations or thoughts about what college should be like? What are they? Where do those expectations come from (family, peers, social media, etc.)?

Answer A

At this point in my life, college is a place for me to grow as an individual and learn traits about myself I have never thought I had. I see college as a chance to step out of my comfort zone to try many new things and make mistakes as this is the only way I will know what is right and wrong. College represents independence and maturity as I step into a new chapter of my life, which gives me a chance to learn about myself and to learn how to manage things on my own such as time and even money. It also means gaining knowledge and many skills I will need in my career through lectures, clubs and hands-on-learning experiences like internships and jobs. College also lets me make my own decisions which gets me ready for the future. I know college will help me discover my interests and passions as I meet new people and join organizations. I think college should be a place to feel challenged and a place that will make you work for your goals which can help build skills that will matter in the future. It should also be a place for someone to feel supported through and through where professors and students can be your mentors. College is a space for giving opportunities to explore different paths and experiences that will help you figure out what truly fits an individual.

Answer B

I think of college as a place to practice making judgments when there is no answer key. In school, it can be tempting to measure success by whether I guessed what the teacher wanted. I hope college asks something harder: what do I believe, what evidence supports it, and what would make me change my mind? I would like to leave with a direction for work, but also with better reasons for choosing that direction.

My picture of college comes partly from the messages surrounding it. Families often emphasize the degree and the employment it might lead to. Social media tends to emphasize the experience: friends, events, and an exciting new identity. Both pictures leave me wondering what an ordinary Tuesday will feel like. I expect much of college to consist of reading, revising work, and showing up even when nothing particularly memorable happens.

I feel confident that being interested in a subject is a useful starting point. I am less certain about how I will respond when interest is not enough and the work becomes difficult. I hope to encounter a course that changes an opinion I took for granted. College should make room for disagreement and questions, including basic ones. Becoming someone who can admit uncertainty without giving up would mean as much to me as choosing the right major.

AI impostor: A / B / Own submission

Confidence: Low / Medium / High

Reason: _____

Round 2 - Rhetoric and philosophy

Question: What are a speaker's two main objectives, and why might measurable facts and moral claims affect persuasion differently? Introduce the distinction and explain how rhetorical techniques fit into it. Here, rhetoric means the use of language to communicate and persuade.

Answer A

A successful speech should leave listeners able to explain the speaker's point and willing to take that point seriously. These are two distinct objectives: communicating a position and persuading an audience. Agreement cannot be assumed merely because the message was understood.

Imagine a proposal to replace a parking lot with a public park. The number of parking spaces that would disappear is a factual question. People can count them and examine the counting method. Whether the park would be a fair use of public land is a moral question. A driver, a nearby resident, and a parent may accept exactly the same numbers while disagreeing about whose needs deserve priority. Moral reasoning therefore requires speakers to explain their standards as well as their evidence. It is not simply a matter of collecting more measurements.

Rhetorical choices connect these arguments to an audience. A diagram could make the land-use proposal easier to understand. A carefully chosen story could show why access to outdoor space matters. Acknowledging the inconvenience to drivers could establish credibility. Such techniques can clarify a sound argument, but they can also conceal weak evidence or unfair assumptions. Persuasive success is consequently different from proof: an audience's approval tells us that a presentation worked, not necessarily that its conclusion was justified.

Answer B

A speaker has two fundamental objectives. The first is to get an intended message across to an audience. This transfer is facilitated by simplicity and clarity, which the speaker aims to achieve "through the structured use of symbols." Speeches use the symbols of language and words. The second aim of the speaker after clarifying his message is to persuade the audience to accept it. It would seem that a speaker is more successful at accomplishing the two objectives when he has validity and truth on his side, but is his rhetoric just as effective without them?

While the truths behind a message influence the effectiveness of a speaker, truths' inherent, vague nature adds a dimension of uncertainty to its effect. Two specific types of truths, quantitative and moral, can have differing effects on rhetorical successes. Quantitative truths deal with those aspects of nature that can be calculated, and the tangible, physical properties of systems. On the other hand, moral truths are more abstract and subject to broad interpretation. In his The History and Theory of Rhetoric: An Introduction, James Herrick conveniently categorizes the symbols and strategies used in persuasive argument into four categories - argument, appeals, arrangement, and aesthetics. With each type of truth, these four simple, yet very prevalent methods of persuasion offer different angles and dimensions from which to approach the truth's influence on a speaker's rhetoric.

AI impostor: A / B / Own submission

Confidence: Low / Medium / High

Reason: _____

Round materials credit: Kevin Liu, Truth and Rhetorical Effectiveness, MIT 21W.747 Rhetoric, Spring 2005, MIT OpenCourseWare. Excerpt selection, formatting, and new question/alternative response are adaptations under CC BY-NC-SA 4.0. Credit applies to the round; the answer mapping is in the instructor key. No endorsement is implied.

Round 3 - Energy and environmental policy

Question: Discuss the environmental benefits and drawbacks of nuclear fission power, and explain how public reactions to accidents can influence support for it. Fission is the splitting of atomic nuclei to release energy. Treat this as a discussion framed in 2006, not an update on today's energy policy.

Answer A

Fission has positive attributes, namely that it doesn't produce greenhouse gases or air pollutants such as carbon monoxide, sulfur dioxides or nitrogen oxides and the fuel supply is abundant. However, the drawbacks outweigh the benefits. Extraction of U-235 results in habitat destruction and radioactive contamination of local ecosystems. The operation of nuclear power plants results in the generation of radioactive waste, the release of radiation, and thermal pollution as the water used to cool the reactor core is dumped into the environment, altering the temperature of lakes and streams by several degrees. Additionally, the use of fissionable material as fuel presents nuclear proliferation risks. The plants are also expensive to build and costly to maintain and decommission once operation has ceased.

Nuclear power had promising beginnings in the 1970s, with orders for power plants reaching almost 45 in 1974. With the growth of the environmental movement in the early 1970s, strong public opposition developed against nuclear power. Media coverage of the malfunctioning cooling system at Three Mile Island in 1979 as well as the explosion of the nuclear reactor at Chernobyl in 1986 resulted in widespread public unease which remains today and accounts for the reluctance of policymakers to consider nuclear power as a viable option.

Answer B

In a 2006 debate, the environmental case for nuclear fission would depend heavily on what would supply the electricity instead. Replacing a coal-fired station with a nuclear plant could reduce the air pollution and carbon dioxide associated with burning coal. A reactor can also generate large amounts of electricity continuously, which makes it useful for meeting steady demand. These benefits matter when evaluating an energy system rather than a single facility in isolation.

The difficult tradeoff is that some environmental responsibilities continue long after electricity has been sold. Used fuel remains radioactive and requires secure management. Mining and fuel preparation disturb land and consume resources, while a plant's cooling needs can place demands on local water supplies. These effects mean that assessing nuclear power requires looking at its whole life cycle, including eventual closure.

Accidents introduce a different kind of concern: people must trust institutions to prevent events they cannot personally monitor. A highly publicized failure can make official assurances less convincing, even at facilities with different designs. Residents may demand stronger oversight, oppose a proposed site, or pressure elected officials to delay approval. Support therefore depends on both technical performance and credible public accountability. Comparing ordinary operating benefits with the possibility of severe harm is ultimately a policy judgment, not a question that electricity output alone can settle.

AI impostor: A / B / Own submission

Confidence: Low / Medium / High

Reason: _____

Round materials credit: Jessica Lynch and Laura Jacox, Why Fusion? The Politics and Policy of Energy in the United States, MIT 22.012 Seminar: Fusion and Plasma Physics, May 11, 2006, MIT OpenCourseWare. Excerpt selection, formatting, and new question/alternative response are adaptations under CC BY-NC-SA 4.0. Credit applies to the round; the answer mapping is in the instructor key. No endorsement is implied.

Round 4 - Psychology

Question: Ice cream sales and crime rates can rise at the same time. Does this mean that buying ice cream causes crime? Explain a more plausible interpretation of this relationship.

Answer A

The two increases do not establish that ice cream purchases cause crime. A correlation describes how measurements vary together; a causal claim says that changing one factor would change the other. Those are different statements. The sales figures alone provide no evidence that preventing a purchase would prevent an offense.

One plausible explanation is a seasonal influence on both measurements. During hotter months, demand for frozen desserts rises. Changes in outdoor activity, gatherings, and the number of people using public spaces may also affect opportunities for certain offenses. Season or weather could therefore create an association even if ice cream itself has no effect on criminal behavior. This is an example of a possible common cause.

To investigate, researchers could compare days with similar temperatures and examine whether the relationship persists. They would also need to consider location, population size, and which crimes were counted. For instance, a busier district could record more sales and more incidents simply because more people are present. None of these possibilities is proved by the original observation. The appropriate conclusion is that the pattern calls for further explanation, with seasonal conditions offering a more plausible starting hypothesis than an effect of eating ice cream.

Answer B

Did you know that as sales in ice cream increase, so does the overall rate of crime? Is it possible that indulging in your favorite flavor of ice cream could send you on a crime spree? Or, after committing crime do you think you might decide to treat yourself to a cone? There is no question that a relationship exists between ice cream and crime (e.g., Harper, 2013), but it would be pretty foolish to decide that one thing actually caused the other to occur.

It is much more likely that both ice cream sales and crime rates are related to the temperature outside. When the temperature is warm, there are lots of people out of their houses, interacting with each other, getting annoyed with one another, and sometimes committing crimes. Also, when it is warm outside, we are more likely to seek a cool treat like ice cream. How do we determine if there is indeed a relationship between two things? And when there is a relationship, how can we discern whether it is attributable to coincidence or causation?

AI impostor: A / B / Own submission

Confidence: Low / Medium / High

Reason: _____

Round source credit: Rose M. Spielman, William J. Jenkins, and Marilyn D. Lovett, OpenStax, Psychology 2e. Access for free at https://openstax.org/books/psychology-2e/pages/1-introduction. This educational adaptation is shared under CC BY-NC-SA 4.0; excerpt selection and new question/answer material are adaptations. No OpenStax endorsement is implied.

Round 5 - Economics

Question: Selena paid $8 for a nonrefundable movie ticket. After 30 minutes, she dislikes the movie, and 90 minutes remain. Should the money already spent be a reason to stay? Explain sunk costs and the cost of staying.

Answer A

Consider the case of Selena, who pays $8 to see a movie, but after watching the film for 30 minutes, she knows that it is truly terrible. Should she stay and watch the rest of the movie because she paid for the ticket, or should she leave? The money she spent is a sunk cost, and unless the theater manager is sympathetic, Selena will not get a refund. However, staying in the movie still means paying an opportunity cost in time. Her choice is whether to spend the next 90 minutes suffering through a cinematic disaster or to do something-anything-else. The lesson of sunk costs is to forget about the money and time that is irretrievably gone and instead to focus on the marginal costs and benefits of current and future options.

For people and firms alike, dealing with sunk costs can be frustrating. It often means admitting an earlier error in judgment. Many firms, for example, find it hard to give up on a new product that is doing poorly because they spent so much money in creating and launching the product. However, the lesson of sunk costs is to ignore them and make decisions based on what will happen in the future.

Answer B

The decision can be organized around one question: which available use of the next 90 minutes offers Selena the greatest value? Her ticket price does not change across those options. If she leaves, the theater keeps the $8. If she stays, the theater still keeps the $8. Because that payment cannot be recovered, it is a sunk cost and should not count as an additional benefit of remaining seated.

Suppose Selena could instead use the time to finish an assignment she would otherwise complete late at night. Staying would mean giving up that opportunity, even though watching the rest of the movie requires no further payment. Economists call the value of the best forgone alternative the opportunity cost. The cost of staying is therefore not necessarily zero just because she already has a ticket.

She should compare that cost with what she expects to gain from finishing the film. Perhaps she still wants to learn how the story ends; perhaps she expects no enjoyment at all. Either assessment concerns the future and is relevant. The thought that she must watch every minute to get her money's worth concerns an irreversible payment. Leaving cannot undo the purchase, but it can prevent a disappointing purchase from also consuming time she would prefer to use elsewhere.

AI impostor: A / B / Own submission

Confidence: Low / Medium / High

Reason: _____

Round source credit: Steven A. Greenlaw and David Shapiro, OpenStax, Principles of Economics 2e. Access for free at https://openstax.org/books/principles-economics-2e/pages/1-introduction. This educational adaptation is shared under CC BY-NC-SA 4.0; excerpt selection and new question/answer material are adaptations. No OpenStax endorsement is implied.

Round 6 - Biology

Question: A population of finches has a different average bill size after several generations. Does that mean that individual birds changed their bills through evolution? Explain the difference between an individual changing during its life and a population evolving.

Answer A

A change in average bill size does not mean that the original finches gradually reshaped their bills and passed those acquired changes to their offspring. The birds measured several generations later are largely different individuals. The central question is whether inherited characteristics have become more or less common in the population.

For example, suppose the starting population contains inherited variation in bill size. If birds with one kind of bill leave more surviving offspring under the prevailing conditions, their characteristics may become more common in later generations. The population's average can shift even though no adult bird changes its own inherited bill characteristics. Natural selection is one process that can produce such a shift; chance and movement of birds between populations can also affect inherited variation.

An individual's development is a different matter. A young bird grows, and its body may respond to nutrition or other conditions during its lifetime. Those changes do not automatically constitute evolution. Indeed, an observed change in average bill size is not sufficient by itself to demonstrate evolutionary change: researchers would need to distinguish inherited differences from environmental effects or differences in the ages of birds sampled. Evolution concerns changes in inherited population characteristics across generations, while development concerns changes within an organism's life.

Answer B

Evolution is the change in a population's genetic composition over time, specifically over generations, resulting from differential reproduction of individuals with certain alleles. Individuals do change over their lifetime, obviously, but this is development and involves changes programmed by the set of genes the individual acquired at birth in coordination with the individual’s environment.

When thinking about the evolution of a characteristic, it is probably best to think about the change of the average value of the characteristic in the population over time. For example, when natural selection leads to bill-size change in medium ground finches in the Galápagos, this does not mean that individual bills on the finches are changing. If one measures the average bill size among all individuals in the population at one time and then measures them in the population several years later, this average value will be different as a result of evolution. Although some individuals may survive from the first time to the second, they will still have the same bill size; however, there will be many new individuals who contribute to the shift in average bill size.

AI impostor: A / B / Own submission

Confidence: Low / Medium / High

Reason: _____

Round source credit: Mary Ann Clark, Matthew Douglas, and Jung Choi, OpenStax, Biology 2e. Access for free at https://openstax.org/books/biology-2e/pages/1-introduction. This educational adaptation is shared under CC BY-NC-SA 4.0; excerpt selection and new question/answer material are adaptations. No OpenStax endorsement is implied.

Round 7 - Sociology

Question: How do formal norms differ from informal norms? Explain how people learn informal norms, and use everyday examples to show how unwritten rules shape behavior.

Answer A

Consider a seminar where the syllabus prohibits plagiarism, but nobody has written down how long a student should speak before allowing someone else a turn. Both expectations influence behavior. The first is a formal norm, explicitly established and backed by an institutional process. The second is an informal norm, maintained largely through shared expectations and social reactions.

People learn informal norms through participation. A new student may notice that classmates pause before responding, recognize that interruptions receive disapproving looks, and adjust accordingly. Someone might also explain the expectation directly. Approval, embarrassment, humor, and exclusion can all communicate what a group considers acceptable without an official penalty being imposed.

Everyday technology provides another example. A group chat may have an unwritten understanding that urgent messages deserve a quick reply while ordinary messages can wait. Sending repeated demands for an immediate response could be considered rude even though it breaks no written rule. In a shared apartment, returning borrowed items promptly may similarly be expected without appearing in the lease.

These expectations are learned rather than universal. Different groups may have different ideas about interruptions, response times, or borrowing. Misunderstandings can occur when someone applies the rules of one setting to another. Informal norms help coordinate daily life, but they can also create pressure to conform.

Answer B

Formal norms are established, written rules. They are behaviors worked out and agreed upon in order to suit and serve the most people. Laws are formal norms, but so are employee manuals, college entrance exam requirements, and “no running” signs at swimming pools. Formal norms are the most specific and clearly stated of the various types of norms, and they are the most strictly enforced. But even formal norms are enforced to varying degrees and are reflected in cultural values.

There are plenty of formal norms, but the list of informal norms-casual behaviors that are generally and widely conformed to-is longer. People learn informal norms by observation, imitation, and general socialization. Some informal norms are taught directly-“Kiss your Aunt Edna” or “Use your napkin”-while others are learned by observation, including observations of the consequences when someone else violates a norm.

But although informal norms define personal interactions, they extend into other systems as well. In the United States, there are informal norms regarding behavior at fast food restaurants. Customers line up to order their food and leave when they are done. They don’t sit down at a table with strangers, sing loudly as they prepare their condiments, or nap in a booth. Most people don’t commit even benign breaches of informal norms.

AI impostor: A / B / Own submission

Confidence: Low / Medium / High

Reason: _____

Round source credit: Heather Griffiths and Nathan Keirns, OpenStax, Introduction to Sociology 2e. Access for free at https://openstax.org/books/introduction-sociology-2e/pages/1-introduction-to-sociology. This educational adaptation is shared under CC BY-NC-SA 4.0; excerpt selection and new question/answer material are adaptations. No OpenStax endorsement is implied.

Round 8 - Astronomy

Question: Why is the changing distance between Earth and the Sun an inadequate explanation of the seasons? Explain how Earth's tilt accounts for opposite seasons in the Northern and Southern Hemispheres.

Answer A

Many people have believed that the seasons were the result of the changing distance between Earth and the Sun. This sounds reasonable at first: it should be colder when Earth is farther from the Sun. But the facts don't bear out this hypothesis. Although Earth's orbit around the Sun is an ellipse, its distance from the Sun varies by only about 3%. That's not enough to cause significant variations in the Sun's heating. To make matters worse for people in North America who hold this hypothesis, Earth is actually closest to the Sun in January, when the Northern Hemisphere is in the middle of winter. And if distance were the governing factor, why would the two hemispheres have opposite seasons? As we shall show, the seasons are actually caused by the 23.5° tilt of Earth's axis.

Note that our axis continues to point the same direction in the sky throughout the year. As Earth travels around the Sun, in June the Northern Hemisphere "leans into" the Sun and is more directly illuminated. In December, the situation is reversed: the Southern Hemisphere leans into the Sun, and the Northern Hemisphere leans away. In September and March, Earth leans "sideways"-neither into the Sun nor away from it-so the two hemispheres are equally favored with sunshine.

Answer B

Seasons depend on the amount of solar energy a location receives over a day. Earth's tilted axis changes that amount in two connected ways: it changes the Sun's height in the sky and the duration of daylight. Both effects must be considered to explain seasonal heating.

When a hemisphere tilts toward the Sun, sunlight generally arrives at a steeper angle. The energy is concentrated over a smaller area of ground than it would be under low-angle illumination. Longer days also give the surface more hours to receive sunlight. Together, these conditions produce summer. The opposite hemisphere has lower-angle sunlight and shorter days, producing winter. About six months later, Earth's position in its orbit makes the other hemisphere face the Sun more directly.

The axis stays pointed in approximately the same direction as Earth completes its orbit; it does not need to flip back and forth. Its tilt is about 23.5 degrees from the direction perpendicular to the orbital plane.

Changes in the Earth-Sun distance can alter the energy reaching Earth, but they act on the planet as a whole. They cannot explain why Australia and Canada experience opposite seasons at the same time. Axial tilt explains that contrast because it changes how incoming sunlight is distributed between the hemispheres.

AI impostor: A / B / Own submission

Confidence: Low / Medium / High

Reason: _____

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